

Assessment Tool-2

CO-1

Concept Mapping

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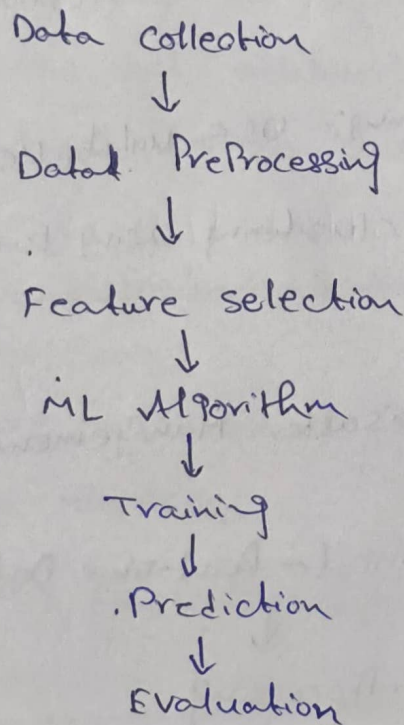
Reg No:- 192525382

Course Code:- ITA0612

Course Name:- Machine Learning.

1) Blocks in a Machine learning system:-

A machine learning system consists of the following major blocks:-



Data collection:- Collect data from sensors, databases or files.

PreProcessing:- Remove missing, duplicate and noisy data.

Feature selection:- Select important features.

Training:- The algorithm learns Patterns from training data.

Prediction:- The trained model Predicts outputs for new data.

Evaluation:- Accuracy and other metrics and used to measure Performance

Supervised learning:- Uses labelled data.

Ex:- Spam / Not Spam classification.

Unsupervised learning:- Uses Unlabelled data

Ex:- Customer clustering using k-means.

2) ML System for Disaster Management:-

Historical + Real-time Data



PreProcessing



Feature selection



Random Forest



Risk Prediction



Early warning.

- Historical rainfall, water level, temperature and seismic data are collected.
 - Missing and noisy values are removed.
 - Important feature are selected.
 - Random Forest can be used because it handles multiple features and nonlinear relationships.
 - Data is divided into training, validation and testing sets.
 - The model is evaluated using accuracy, Precision, recall and F1-score.
 - If high risk is Predicted, an alert is sent to authorities and the Public.
 - This help in early evacuation and emergency response.
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3) Find-S Algorithm - Play Tennis:-

Positive examples:-

1. Overcast, Hot, High, weak
2. Rainy, mild, High weak.
3. Rainy, cool, Normal weak
4. overcast, mild, Normal, strong.

Hypothesis updates:-

$[S_0 = \langle \emptyset, \emptyset, \emptyset, \emptyset \rangle]$.

After Example 3:-

$[S_1 = \langle \text{overcast}, \text{Hot}, \text{High}, \text{weak} \rangle]$

After Example 4:-

$[S_2 = \langle ?, ?, \text{High}, \text{weak} \rangle]$

After Example 5:-

$[S_3 = \langle ?, ?, ?, \text{weak} \rangle]$

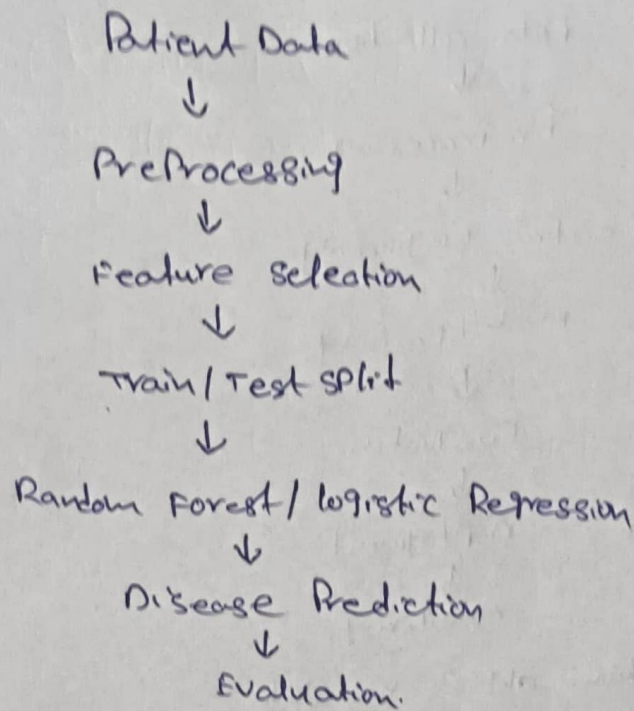
After Example 7:-

$[S_4 = \langle ?, ?, ?, ? \rangle]$

Final Answer:-

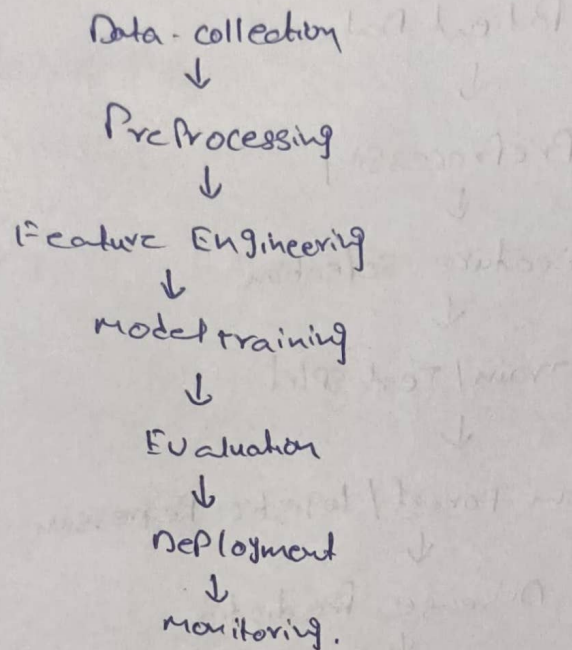
$\boxed{S = \langle ?, ?, ?, ? \rangle}$

10) ML System for Health Analysis:-



- Patient data such as age, blood Pressure, glucose and lifestyle are collected.
- missing values and duplicate records are handled.
- Categorical data is encoded.
- Important features are selected.
- Logistic Regression is suitable for binary disease Prediction.
- Random Forest can capture nonlinear relationships.
- The dataset is divided into training and testing sets.
- Accuracy, Precision, Recall, F1-score and ROC-AUC are used for evaluation.

11) Machine Learning Pipeline:-



- Data Collection:- obtains raw data
- Pre Processing:- Cleans and transforms the data.
- Feature Engineering:- Creates useful features.
- Training:- ML algorithm learns Patterns.
- Evaluation:- Measures model Performance
- Deployment:- Uses the model in a real application.
- Monitoring:- Continuously checks model Performance.

Training Error vs Generalization Error:-

Training Error:- Error on the training data.

Generalization Error on unseen data.

Example:-

[Training | Accuracy = 98%.

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[Testing | Accuracy = 70%.

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